

SYSTEM ARCHITECTURE & COMPLETE SPECIFICATION

LAYERED-HIERARCHICAL-CATEGORICAL:

Hierarchical Entity Extension:

Geographical Hierarchy

- Parent-child relationships from Geolocation → District → City → Market
- Enables precise location-based queries and administration

User Hierarchy

- Inheritance Model: Users → Admin → Member/Vendor → Supplier
- Admin User Hierarchy: Super Admin → URA → City Admin → Market Master → Pseudo Market Master → Counter

User Inheritance Model

- Admin Branch: System management roles with increasing specialization
- Stakeholder Branch: Market participants with commercial interests
- Single User table with specialized extensions

Pseudo Admin Specializations

1. Revenue Collector - Payment processing & revenue reporting
2. Health Inspector - Sanitation & compliance inspections
3. Security Admin - CCTV monitoring & incident response
4. Gate Counter Staff - Vehicle entry/exit logging
5. Stock Counter Staff - Inventory management & tracking
6. VAT Payment Counter - Tax collection & processing

MARKET TYPES:

By Scale:

- **By Size:**
 - Large-scale (hundreds of vendors, thousands of visitors)
 - Medium-scale (dozens to hundreds of vendors)
 - Small-scale (few vendors, limited visitors)
- **By Duration:**
 - Permanent markets (year-round)
 - Temporary markets (weekly, seasonal)

- One-time event markets

By Ownership:

- Government-owned & Private markets
- Public-private partnership markets

By Specialization:

- General merchandise markets
- Fresh produce markets
- Clothing and textile markets
- Electronics markets
- Specialty markets (crafts, antiques)

Universal Category System

- Single *Category* table with *category_type* field
- Polymorphic *CategoryAssignment* for flexible entity tagging
- Supports hierarchical categories through *parent_id*



ARCHITECTURAL HIGHLIGHTS:

Core Business Domains:

- Hierarchical Location Management: (City→Area→Market→Gate→Level→Shop)
- User Management: Unified user system with RBAC & KYC
- Vendor Ecosystem: Complete vendor lifecycle from registration to operations
- Product Flow: End-to-end product tracking from gate intake to point of sale
- Financial System: Comprehensive tax and payment management
- Inventory Control: Real-time stock management with audit trails

Scalability & Extensibility:

Flexible Relationships

- Relation table for generic entity-to-entity connections
- Support for many-to-many relationships between any entities

Supporting Entities

- Audit logging with full change tracking
- Digital assets & Document categorization
- Notification system with multiple channels
- QR/Token system for various purposes

- Form management & categorization

Business Impacts:

For Local Government:

- Centralized Management, Revenue Optimization & Operational Efficiency
- Digitalized market operations and vendor management
- Compliance: Regulatory requirement enforcement and audit trails

For Vendors/Shop Owners:

- Business Management Control, Financial Transparency with Clear payment tracking and tax calculations

MODULE-WISE COMPLETE FEATURES LIST

Module 1: User Management & Authentication

- Multi-level RBAC with hierarchical permissions
 - TOTP two-factor authentication
 - Session management & device tracking
 - User Registration Link Management
- User profile management & KYC
 - User activity monitoring & Logging
 - Customer profile management
 - Password policy enforcement
- Account lockout action mechanism
- Bulk user operations

Module 2: Market & Shop Management

- Market configuration & Level-shop management
 - Market facility oversight
 - Multi-market hierarchy support
 - Visual floor plan management
 - Market performance analytics
- Shop allocation and rental management
 - Shop vacancy & status tracking
 - Shop rent billing system
 - Rental rate management

- Facility maintenance scheduling

Module 3: Notification & Communication System

- Multi-channel delivery (Email, SMS, Push, In-App)
- Template management with dynamic variables
- Delivery tracking with read receipts
- Subscription management
- Priority-based notification system
- Scheduled notifications
- Bulk messaging capabilities
- Notification analytics
- Quiet hours management
- Cross-platform compatibility

Module 4: Document Management System

- Multi-format document support
- Version control system
- Advanced access control
- OCR text extraction
- Full-text search capabilities
- Automated watermarking
- Document compression
- Retention policy management
- Bulk document operations
- Document sharing and collaboration

Module 5: QR Code Engine & Token Management

- Dynamic QR code generation
- Real-time validation system
- Batch QR code creation
- Scan analytics and tracking
- Anti-tampering security
- Template-based branding
- Token expiration management
- Blacklist functionality

- Bulk token operations
- Digital signature support

Module 6: Gate Management & Logistics

- Vehicle entry/exit tracking
- License plate recognition
- Real-time token validation
- CCTV integration
- Traffic analytics
- Blacklist management
- Emergency protocols
- Gate scheduling
- Vehicle type management
- Driver profile tracking

Module 7: Revenue Management System

- Automated billing system
- Multi-payment gateway integration
- VAT/tax calculation engine
- Invoice generation
- Payment reconciliation
- Revenue reporting
- Tax configuration management
- Refund processing
- Payment plan management
- Financial analytics

Module 8: Inventory & Stock Management

- Real-time stock tracking
- Supplier management portal
- Barcode/RFID integration
- Stock alert system
- Inventory valuation
- Batch tracking
- Stock movement history

- Warehouse management
- Purchase order management
- Stock reconciliation

Module 9: Point of Sale (POS) System

- Multi-payment method support
- Offline capability with cloud sync
- Receipt generation
- Inventory synchronization
- Customer loyalty programs
- Tax computation
- Discount management
- Sales analytics
-

Module 10: Payment Gateway Integration

- Unified payment processing
- Multiple gateway support
- Transaction monitoring
- Reconciliation system
- Webhook handling
- Fee calculation
- Refund management
- Gateway health monitoring
- Security compliance
- Settlement tracking

Module 12: IoT & CCTV Management

- Device management system
 - Alert generation
 - Video storage management
 - Maintenance scheduling
 - Device health monitoring
 - Access control integration
- Live video streaming

- Motion detection
- Face recognition
- Object tracking

Module 13: Analytics & Reporting

- Real-time dashboard & Custom report builder
- Data visualization & Predictive analytics
- Automated reporting & Export functionality
- Data caching & Performance monitoring
- KPI tracking & Trend analysis

Extensible Modules & Features Identified

Supporting Functionalities & Features

1. Search & Filtering Functionality
 - a. Search Bar & Filtering with multiple criteria
2. Backup & Data Recovery Module:
 - a. Data backup scheduling
 - b. Data validation and cleanup
3. System Administration Panel:
 - a. System performance metrics
 - b. Real-time system health dashboard
 - c. API & Gateway Management
4. Compliance & Public Health Monitoring
5. Data/Doc Import/Export & Management
 - a. Data/Doc Processing & Management
6. Bulk Operations Management:
 - a. Bulk user creation/management
 - b. Batch processing functionality
7. Disaster & Risk Management
 - a. Risk assessment & Incident reporting system
 - b. Loss Calculation, Compensation & claim processing
 - c. Disaster Preparation, Recovery & Alert System

FUNCTIONAL REQUIREMENTS

High-Priority Features

FR-001: Link based User Registration Process

- Link Generation & Management
- User Onboarding & Link Status
- Auth, Validation, Verification, Session
- KYC, RBAC, MFA, DOC Management

FR-002: Multi-Role Authentication

- Secure login with role-based permissions
- Password recovery mechanism
- Session timeout management

FR-003: Gate Management System

- Vehicle entry/exit tracking
- Token generation and validation
- Customer/supplier registration

FR-004: Stock Processing

- Purchase and delivery stock entry
- VAT calculation automation
- Inventory level tracking

FR-005: Revenue Collection

- VAT payment processing
- Rent collection management
- Financial reporting

FR-006: Market Administration

- Shop allocation and management
- Member/vendor services
- Market configuration

SUCCESS METRICS

Key Performance Indicators

- Revenue collection efficiency
- Transaction processing time
- User adoption rates
- System uptime and reliability
- Customer satisfaction scores